Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [20EE34]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ELECTROMAGNETIC FIELD THEORY

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	If $\vec{B} = y\vec{a}_x + (x+z)\,\vec{a}_y$ and a point Q is located at (-2,6,3), express (1) the point Q in cylindrical and spherical coordinates, (2) $\vec{B}$ in spherical coordinates	**14**
2.	**a.**	By means of Guass's Law, determine the electric field intensity at a point P distant 'h' m from an infinite line of uniform charge ρ_l C/m.	**07**
	b.	State and derive electric boundary conditions for a dielectric to dielectric medium and a conductor to dielectric medium.	**07**
3.	**a.**	Derive the expression for magnetic field intensity inside and outside a coaxial conductor of inner radius "a" outer radius "b" and carrying a current of I amperes in the inner and outer conductors.	**14**
		OR	
4.	**a.**	State Ampere's Circuital law.	**06**
	b.	Derive the expression for inductance per unit length of a long solenoid of N turns and having a length 'l' m carrying a current of I amperes.	**08**
5.	**a.**	Derive Maxwell's equation both in integral form.	**06**
	b.	Derive an expression for torque in a rectangular loop which is carrying a current of 'I' amperes and is situated in a uniform magnetic field 'B' Wb/m2	**08**
		OR	
6.	**a.**	Distinguish between transformer emf and motional emf	**04**
	b.	State Ohm's law for magnetic circuits and also differentiate circuit theory and field theory.	**10**
7.	**a.**	Define Poynting vector. Explain its significance.	**07**
	b.	List out the properties of a uniform plane wave.	**07**
		OR	
8.	**a.**	Derive the wave equation in free space by assuming all required conditions and constants.	**14**